At about the time that Ginzburg was making this discovery, only a few kilometers away in Moscow Zel'dovich's team—with Igor Novikov and Andrei Doroshkevich taking the lead—began to ask themselves, "Since a round star produces a round hole when it implodes, will a deformed star produce a deformed hole?" As an extreme example, will a square star produce a square hole? (Figure 7.3b). To compute the implosion of a hypothetical square star would be exceedingly difficult, so Doroshkevich, Novikov, and Zel'dovich focused on an easier example: When a nearly spherical star implodes with a tiny mountain sticking out of its surface, will the hole it forms have a mountain-like protrusion on its horizon? By asking about nearly spherical stars with tiny mountains, the Zel'dovich team could simplify their calculations greatly; they could use mathematical techniques called *perturbation methods* that John Wheeler and a postdoc, Tullio Regge, had pioneered a few years earlier. These perturbation methods, which are explained a bit in Box 7.1, were carefully designed for the study of any small "perturbation" (any small disturbance) of an otherwise spherical situation. The gravitational distortion due to a tiny mountain on the Zel'dovich team's star was just such a perturbation.

Doroshkevich, Novikov, and Zel'dovich simplified their calculation still further by the same trick that Oppenheimer and Ginzburg used: Instead of simulating the full, dynamical implosion of a mountain-endowed star, they examined only a sequence of static, mountainous stars, each one more compact than the one before. With this trick, and with perturbation techniques, and with intensive give-and-take amongst themselves, Doroshkevich, Novikov, and Zel'dovich quickly discovered a remarkable result: When a static, mountain-endowed star is small enough to form a black hole around itself, the hole's horizon must be precisely round, with no protrusion (Figure 7.3c).

Similarly, it was tempting to conjecture that if an imploding square star were to form a black hole, its horizon would also be round, not square (Figure 7.3b). If this conjecture was correct, then a black hole should bear no evidence whatsoever of whether the star that created it was square, or round, or mountain-endowed, and also (according to Ginzburg) no evidence of whether the star was magnetized or free of magnetism.

A few years later, as this conjecture was gradually turning out to be correct, John Wheeler invented a pithy phrase to describe it: *A black hole*